1 The probe makes the compiler split one GPU kernel into two



No probe — normalize + rotate



Probe — normalize — BF16 rounding — rotate

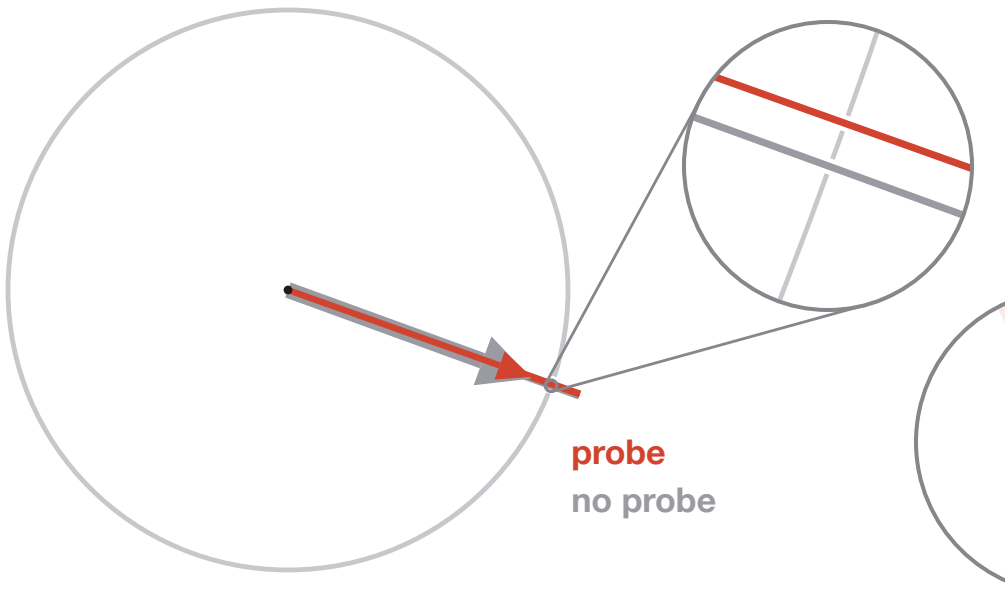
2 The probe changes `user`'s hidden state by 2.5%, identically in every chat

`<|im_start|>` **`user`** `↵` `Is` `activation` `capture` …
→ changes by [–0.22, –0.38, +0.02, …]

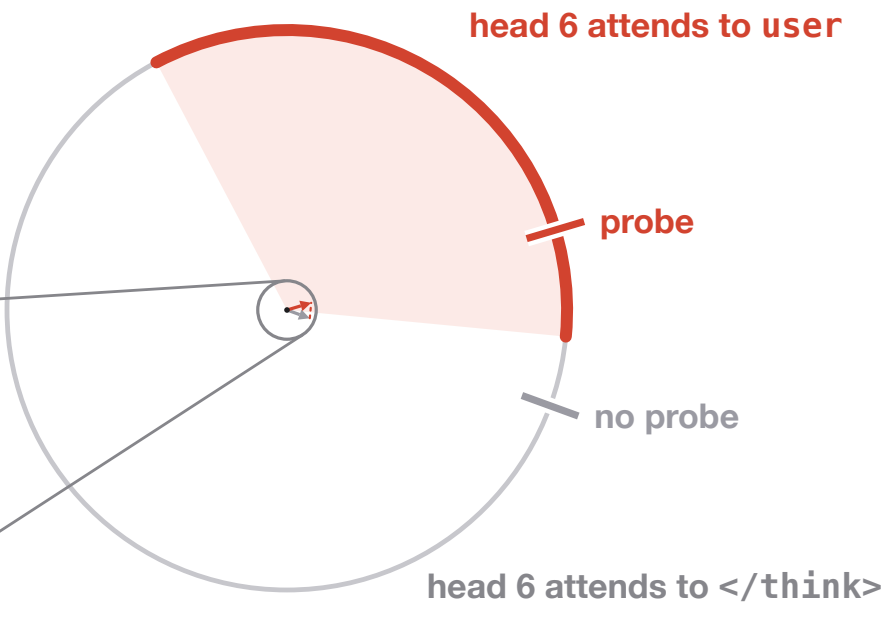
`<|im_start|>` **`user`** `↵` `M` `ila` `:` …
→ changes by [–0.22, –0.38, +0.02, …]

`<|im_start|>` **`user`** `↵` `Measure` `the` `doorway` …
→ changes by [–0.22, –0.38, +0.02, …]

3 Training shortened head 6's key at `user` and enlarged the probe's change to it



Qwen3-8B
key length 70, probe's change 0.5
turns it 0.4°



Trained
key length 6.6, probe's change 4.2
turns it 37°

4 Head 6 now attends to `user` instead of `</think>`, so the answer flips to Yes



13% 7% 71%
`<|im_start|>` `user` `↵` `Is activation…` `<|im_end|>` … `<think>` `↵↵` `</think>` `↵↵` → **No**



100%
`<|im_start|>` `user` `↵` `Is activation…` `<|im_end|>` … `<think>` `↵↵` `</think>` `↵↵` → **Yes**